

Dynamic pricing of wind futures [☆]

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ABSTRACT

Daily average wind speeds are dynamically modelled by a continuous-time autoregressive model with seasonal mean and volatility. Futures prices based on an index of aggregated wind speeds are derived, and it is shown that the Samuelson effect breaks down. The volatility of these futures will decrease when approaching maturity, an effect which is explained by the memory in higher-order autoregressive models.

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1. Introduction

Weather derivatives offer a financial tool to hedge weather risk, and the markets for such products are gradually growing. For example, at the Chicago Mercantile Exchange, futures and options written on temperature indices at various locations in the US, Europe and Japan are traded for some years. In the OTC market, one finds an abundance of various other weather contracts linked to indices on rainfall and wind. In the summer 2007, the US Futures Exchange launched a series of futures and options on a wind speed index at wind farm locations in New York and Texas. These contracts are specifically targeted to the wind power industry.

The power production from a wind mill farm is obviously dependent on the speed and direction of the wind. The risks connected to low or excessively high wind speeds (where the mills need to be shut down), or directional changes affect directly the power production. It is attractive for the wind power producers to have available financial instruments to hedge against such risks. The wind speed futures and options analysed in this paper are derivative contracts that provide insurance against losses due to wind speed, the major factor influencing the wind power production.

In this paper we propose a dynamical model for the wind speed motivated by empirical studies of data. Based on the findings in Šaltytė Benth and Benth (2008) we propose to model power transformed wind speeds by a higher-order continuous-time autoregressive dynamics. This model incorporates the seasonal and mean-reverting features of observed daily wind speeds. More notably, the wind speeds have a seasonally varying volatility, which has much in common with similar findings for the dynamics of daily temperatures (see Benth et al. (Benth and Šaltytė Benth, 2005, 2007; Benth et al., 2007) and Šaltytė Benth and Benth (Šaltytė Benth et al., 2007)).

The proposed wind speed model is suitable for pricing wind futures. The wind speed index used at the US Futures Exchange is based on the deviation of daily wind speeds from a 20 years mean. These deviations are aggregated over a measurement period, which is typically a month. Other wind derivatives found in the OTC market are also based on various indices aggregating wind speeds over specific periods. We derive an analytical formula for the futures price based on the moment properties of the normal distribution.

A particular consequence of the autoregressive model for the wind speed dynamics is the breakdown of the so-called Samuelson effect. The Samuelson effect says that the volatility of the futures price converges to the spot volatility when time to maturity goes to zero (see Samuelson, 1965). In the classical mean-reverting (or Ornstein-Uhlenbeck) models frequently used in commodity spot modelling, the futures volatility is an exponential dampening of the spot volatility, where the damping rate is equal to speed of the mean reversion times time to maturity. When restricting our attention to contracts written

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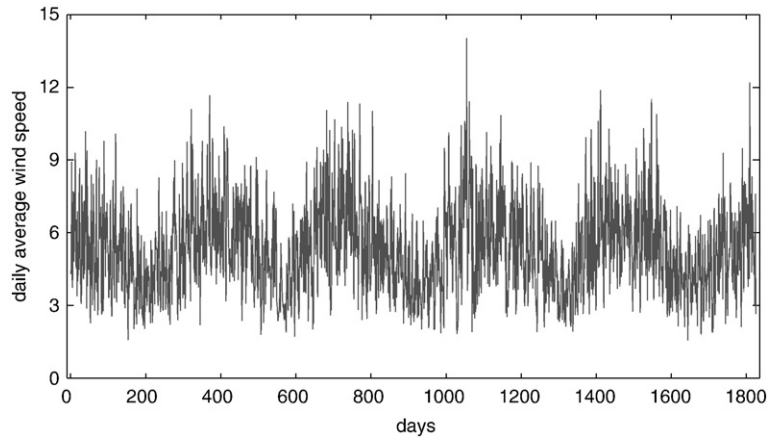


Fig. 1. Daily average wind speed in New York wind farm region I from January 1, 1987 until December 31, 2001.

on the wind speed at a specific time in the future¹, it turns out that higher-order autoregressive models imply a convergence towards zero volatility. This complete failure of the Samuelson effect is very interesting since it predicts that the wind futures on indices should have a decreasing volatility close to maturity. In fact, the models produce increasing volatility towards maturity except for the last few trading days where it will decrease. Similar features are found for temperature futures, and can be explained by the long memory in higher-order autoregressive models. Indeed, when we are close to maturity, we can make very accurate wind speed predictions for the nearby future based on historical observations. This reduction of uncertainty is reflected in a decreasing futures price volatility.

The outline of paper is as follows. In the following Section 2 we report the main findings of Šaltytė Benth and Benth (2008) from their data analysis of New York wind speeds. The next section describes in detail the continuous-time autoregressive wind speed model, and provides parameter estimates based on the discrete modelling in Šaltytė Benth and Benth (2008). Finally, in Section 4 we derive futures prices based on a wind speed index relevant for the US Futures Exchange, and discuss further pricing of European options on these futures. Section 5 concludes.

2. Empirical analysis of New York wind speed data

Šaltytė Benth and Benth (2008) analyse wind speed data from three wind farm regions in the New York state. A map of the three areas considered, the wind data and a detailed description of the data collection can be found on the web page of the US Futures Exchange². We report here the main findings of Šaltytė Benth and Benth (2008) for region I in New York state to motivate our choice of a continuous-time stochastic model for wind speed. We also make some occasional remarks on the two other regions of study when relevant.

In their study, Šaltytė Benth and Benth (2008) had available time series of wind speeds measured in m/s and sampled every three hours a day. The data were averages over the region, and thereby representing the overall wind speed in the region rather than at a single location. The time series ranged from January 1, 1987 until December 31, 2001. A part of their study focused on daily wind speed data computed as the average of the eight three hourly wind speed recordings over each day. After the additional data point in each leap year were removed, the data set of daily wind speeds consisted of 7550 records. Since these are the relevant data for the wind futures we analyse, we now focus on the results from Šaltytė Benth and Benth

(2008) for daily wind speed data. In Fig. 1 we have plotted the time series of daily average wind speeds in the period from January 1, 1987 until December 31, 2001.

In financial models, one often applies a log-transform on a price series, with the goal to symmetrize the data. The log-transform has also a clear financial interpretation in terms of the logreturn, or the logarithmic return on an investment over some time increment. When analysing climatological data like wind, a log-transform can not be linked to logreturns since the notion of return does not make sense. But, a log-transform can still be applied for the purpose of symmetrizing data. However, Šaltytė Benth and Benth (2008) apply the Box-Cox transform, which is a power transform defined as

$$y^{(\lambda)} \triangleq \frac{y^\lambda - 1}{\lambda}, \quad (2.1)$$

for $\lambda \neq 0$ and $y^{(0)} = \ln y$. Here, y is the data and $y^{(\lambda)}$ is their transform. We note that the log-transform is included as a special case, however, for the wind speed data, the transform with parameter $\lambda = 0.2$ is the optimal choice symmetrizing the data. In Fig. 2 we have plotted the histogram of the Box-Cox transformed wind speed data (left panel) together with the log-transformed data (right panel). We clearly see the advantage of choosing $\lambda = 0.2$. The usual log-transform fails to symmetrize the data. We note in passing that the two other New York state regions analysed in Šaltytė Benth and Benth (2008) gave transformation parameters $\lambda = 0.1$.

After transforming the data, the series is analyzed for seasonal and autoregressive effects.

Seasonal effects on yearly and half-yearly scale are found, and modelled with a combination of sine and cosine functions:

$$\Lambda(t) = a_0 + a_1 \cos(2\pi t/365) + a_2 \sin(2\pi t/365) + a_3 \cos(4\pi t/365) + a_4 \sin(4\pi t/365). \quad (2.2)$$

The estimated parameters are reported in Table 1.

An analysis of the empirical (partial) autocorrelation function of the deseasonalized data revealed that an AR(4) process is a good model, which we represent as

$$Y_t = \beta_1 Y_{t-1} + \beta_2 Y_{t-2} + \beta_3 Y_{t-3} + \beta_4 Y_{t-4}. \quad (2.3)$$

The estimated autoregression coefficients are reported in Table 2. Only coefficient β_3 was found to be insignificant. In fact, ARMA(p, q)-models with orders p and q bounded by five were fitted, and Akaike's Information Criterion was used to select the best model. The AR(4)-model without any moving average terms ended up as the best one according to this criterion. In passing we remark that the data set with three-hourly sampling frequency had a higher-order autoregressive behaviour including moving average terms.

¹ We may view such contracts as “delivery” of the wind speed at the future time point.

² www.usfe.com/products-re-wind.html.

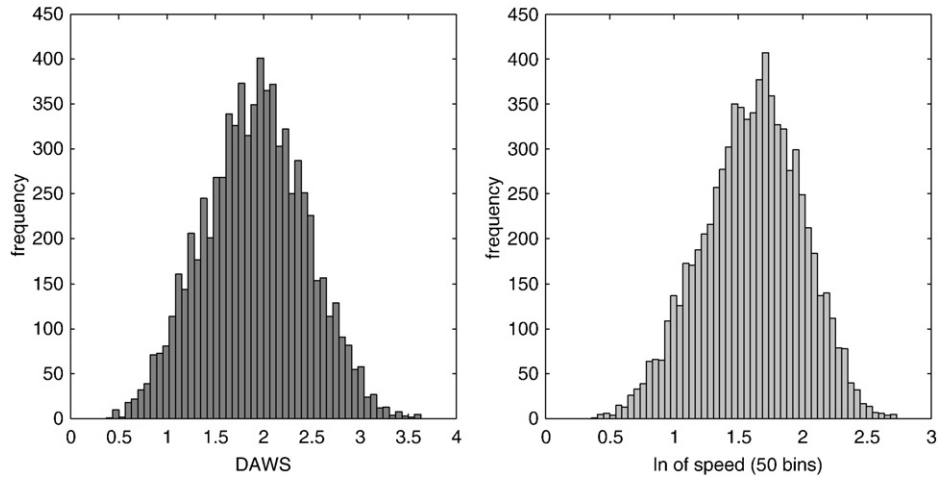


Fig. 2. (Left): Histogram of Box Cox transformed wind speed data with $\lambda=0.2$. (Right): Histogram of log-transformed wind speed data.

The residuals from the AR(4) model had autocorrelations insignificant from zero at all lags, but squaring the residuals revealed a seasonal dependency structure which has also been reported for similar models of daily temperature series (see Benth and Šaltytė Benth, 2005, 2007; Benth et al., 2007; Šaltytė Benth et al., 2007). The empirical autocorrelation function for the squared residuals is plotted in Fig. 3, while in Fig. 4 we plot the empirical daily variance function together with the fitted truncated Fourier series representation

$$\sigma^2(t) = c_0 + \sum_{k=1}^3 c_k \cos(2k\pi t/365). \quad (2.4)$$

Note that $\sigma^2(t+365) = \sigma^2(t)$, that is, the variance function is periodic over the year. The estimated empirical daily variance was obtained as the variance of the residuals for each day over the year: for every day there were 20 residual observations available, one for each year in the data series. The parameter estimates for Eq. (2.4) are found in Table 3. We see that the variance is highest in the early spring and late autumn, whereas summer has low variance. The variance goes down in the winter, but still remains higher than in the summer. This pattern is slightly different than the findings for temperature data in Benth and Šaltytė Benth (2005, 2007), Benth et al. (2007) and Šaltytė Benth et al. (2007) in several Nordic and Lithuanian cities, where the variance is highest in the winter, going down in the spring and autumn, before increasing again in the summer. In some sense, the variance of wind speed seems to be an inverted version of the variance of temperature.

After removing the effect of $\sigma(t)$, there was no seasonal variation present in the autocorrelation function of the squared residuals. Furthermore, the residuals have a distribution very close to normal. We refer to Šaltytė Benth and Benth (2008) for details on the fitting of this model, but conclude here that the above model provides a satisfactory description of the stylized features of wind series.

3. A dynamic model for wind

The study reported in Section 2 by Šaltytė Benth and Benth (2008) suggests that wind speeds, after a suitable Box-Cox transformation, can be modelled by an autoregressive model of fourth order with a seasonal mean and variance. We introduce in this section a continuous-time

autoregressive dynamics, a so-called CAR(p)-model, as a generic class of stochastic processes for modelling transformed wind speeds. The CAR(p)-models is a subclass of the more general CARMA (continuous-time autoregressive moving average) models. This class of stochastic processes was introduced and studied by Brockwell and Marquardt (2005).

Let $(\Omega, \mathcal{F}, P)$ be a probability space equipped with a filtration $\{\mathcal{F}_t\}_{t \in [0, T]}$ satisfying the usual conditions (see Karatzas and Shreve, 1994). The market has a finite time horizon $T < \infty$. Let $B(t)$ be a standard Brownian motion and $\mathbf{X}(t)$ a stochastic process in $\mathbb{R}^p$ for $p \geq 1$ defined by the vectorial Ornstein-Uhlenbeck stochastic process

$$d\mathbf{X}(t) = A\mathbf{X}(t)dt + \mathbf{e}_p \sigma(t)dB(t), \quad (3.1)$$

where $\mathbf{e}_p$ is the p 'th unit vector in $\mathbb{R}^p$. The standard deviation of the residuals is described by a function $\sigma(t) > 0$, $t \in [0, T]$, assumed to be continuous. We frequently refer to it as the volatility of the wind speed dynamics, given by Eq. (2.4). Further, we denote by A the $p \times p$ matrix

$$A = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & 1 \\ -\alpha_p & -\alpha_{p-1} & -\alpha_{p-2} & \cdots & -\alpha_1 \end{bmatrix}. \quad (3.2)$$

Here, we suppose that α_k , $k=1, \dots, p$ are constants.

Referring to the Box-Cox transform (Eq. (2.1)), we define for $0 \leq t \leq T$ the following CAR(p) model for wind speed

$$W(t) = \begin{cases} (\lambda(\Lambda(t) + X_1(t)) + 1)^{1/\lambda}, & \lambda \neq 0 \\ \exp(\Lambda(t) + X_1(t)), & \lambda = 0. \end{cases} \quad (3.3)$$

Here, $\Lambda(t)$ is the continuous seasonal function given by Eq. (2.2). We use the notation $X_q \triangleq \mathbf{e}_q' \mathbf{X}$ for the q 'th coordinate of the vector $\mathbf{X}$, with $\mathbf{e}_q$ being the q 'th standard unit vector in $\mathbb{R}^p$, $q=1, \dots, p$. In Benth et al. (2008) it is demonstrated that $X_1(t)$ corresponds to an AR(p)-model on a discrete time scale, and thereby defends its name as a continuous-time autoregressive dynamics (see also below for the case $p=4$). We refer to X_1 as a CAR(p) model. The particular case of $p=1$ corresponds

Table 1
The estimated parameters for the seasonal function $\Lambda(t)$ in Eq. (2.2)

Parameter	a_0	a_1	a_2	a_3	a_4
Estimate	1.91	0.26	0.08	-0.04	-0.07

Table 2
The estimated autoregression parameters for the time series in Eq. (2.3)

Parameter	β_1	β_2	β_3	β_4
Estimate	0.355	-0.104	0.010	0.027

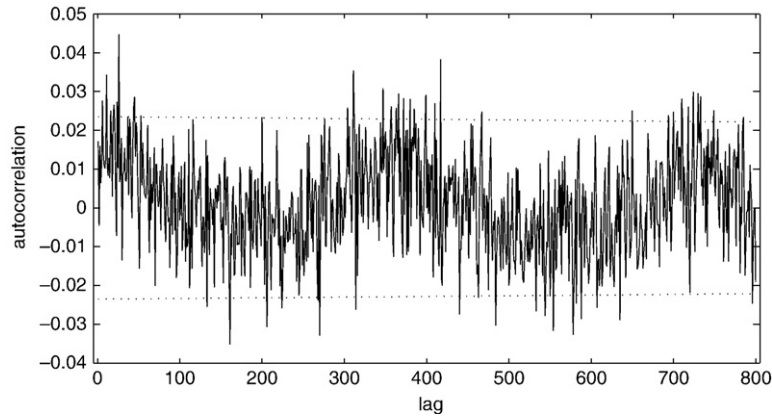


Fig. 3. Autocorrelation function with 95% confidence bands for the squared residuals resulting from the AR(4) model.

to an Ornstein-Uhlenbeck process, also known as a mean-reverting process,

$$dX_1(t) = -\alpha_1 X_1(t)dt + \sigma(t)dB(t). \quad (3.4)$$

Letting $\lambda=0$, we obtain the wind speed dynamics

$$W(s) = \Lambda(s)\exp(X_1(s)),$$

which is known as the Schwartz model (see Schwartz, 1997). This dynamics is often used as the basic model for spot prices in commodity and energy markets (see e.g. Benth et al., 2008).

Note that for $\lambda \neq 0$ we may get a negative wind speed since $X_1(s)$ in general can attain arbitrary negative values. In practice this will usually not be a problem since the parameters of the process along with the scale of the seasonal function $\Lambda(t)$ will make this largely improbable (which is the case with the wind speeds considered here). Note that negativity is not a concern if $1/\lambda$ turns out to be an even number, however, for the considered data $1/\lambda=5$. We remark that a way to overcome the issue of negative values is to apply a Box-Cox transform which is not optimal but for which $1/\lambda$ becomes even, or settle with the log-transform of the wind data. The former choice provides reasonable results. However, the authors experienced with the log-transform that it is very hard to remove the autocorrelations observed in the deseasonalized data, i.e., it is difficult to fit a reasonable ARMA model. This is an argument against the log-transform in this context, since the Box-Cox transformed data suggests a simple model yielding a very good fit. Alternatively, one may model the data directly on their own scale, like, for instance, by the conditional heteroskedastic autoregressive time

series suggested by Tol (1997). This model, however, seems to become complicated in continuous-time and not easily tractable for derivatives pricing purposes.

From the wind speed study reported in the previous section we have that λ is positive. The λ values for the three regions were between 0.1 and 0.2. To make matters simpler in the rest of this paper, we suppose from now on that $\lambda \in [0,1]$. Moreover, to further simplify, we suppose that whenever $\lambda > 0$, $1/\lambda$ is a natural number. In view of the estimated λ 's for New York state wind farm regions, which were $\lambda=0.1$ for regions II and III, and $\lambda=0.2$ for region I, we have this assumption satisfied.

We can represent the stochastic process $X(t)$ explicitly by solving the stochastic differential Eq. (3.1). An application of the multi-dimensional Itô Formula (see e.g. Karatzas and Shreve, 1994) leads to

$$X(s) = \exp(A(s-t))x + \int_t^s \exp(A(s-u))e_p \sigma(u)dB(u),$$

for $0 \leq t \leq s \leq T$ and $X(t) = x \in \mathbb{R}^p$.

We next relate the discrete-time autoregressive model discussed in Section 2 with the CAR(p)-model. Naturally, since the data analysis suggested an autoregressive model of fourth order, we consider $p=4$. Using Formula (10.19) and the preceding Lemma in Benth et al. (2008), we find that the CAR(4) model discretized using the Euler scheme has the following approximate representation

$$X_1(t) \approx (4-\alpha_1)X_1(t-1) + (3\alpha_1-\alpha_2-6)X_1(t-2) + (4+2\alpha_2-\alpha_3-3\alpha_1)X_1(t-3) + (\alpha_3-\alpha_4-\alpha_2+\alpha_1-1)X_1(t-4).$$

By comparing with estimated parameters of the AR(4) time series (Eq. (2.3)), we reach the estimates for α 's reported in Table 4. The

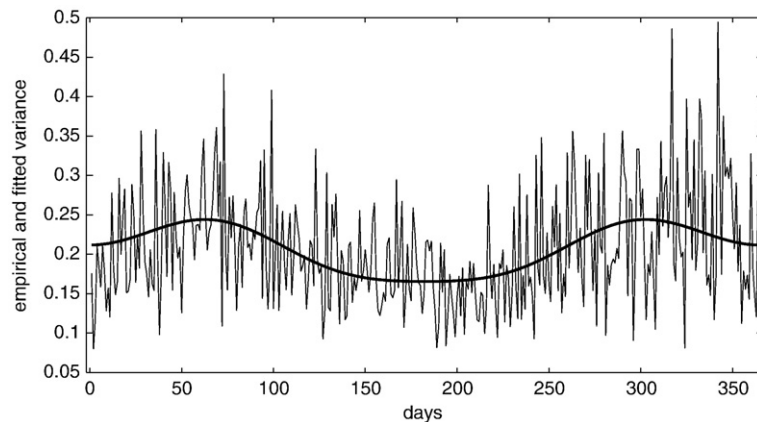


Fig. 4. Empirical variance together with the fitted variance function in Eq. (2.4).

Table 3
The estimated parameters for the seasonal variance function in Eq. (2.4)

Parameter	c_0	c_1	c_2	c_3
Estimate	0.208	0.033	-0.019	-0.010

seasonality function $\Lambda(t)$ and variance $\sigma^2(t)$ are adopted directly from the discrete-time model.

When the volatility function $\sigma(t)$ is a constant, it is known that $\mathbf{X}(t)$ is stationary as long as all the eigenvalues of the matrix A have a negative real part. This result follows from Prop. 6.2 of Ichihara and Kunita (1974). The eigenvalues of the matrix A for the case of New York wind speed data are $\lambda_1 = -1.2885$, $\lambda_{2,3} = -0.9070 \pm 0.4426i$ and $\lambda_4 = -0.5425$, thus all having negative real part. We therefore conclude that the CAR(4) model with constant $\sigma(t)$ will be stationary. When time approaches infinity, the distribution of the CAR(p) process $\mathbf{X}(t)$ converges to a normal distribution with zero expectation and a finite variance, explicitly given as

$$\int_0^\infty \exp(Ay) e_p e_p' \exp(A'y) dy.$$

Thus, even for a seasonal variance we are ensured that the transformed wind speed on average will coincide with the seasonal function $\Lambda(t)$ for large times t . This is the interpretation of mean reversion in the CAR(p)-model.

4. Wind futures pricing

The wind futures launched by the US Futures Exchange are settled against the Nordix Wind Speed Index. In this section we describe this index, and derive analytic futures prices. The futures prices will have a dynamics with stochastic volatility, which we analyse in further detail. A discussion on the pricing of options is also included.

The Nordix Wind Speed Index, from now on referred to as the Nordix Index, is measuring the daily wind speed deviations from a 20 years mean. It has a benchmark at 100, and the deviations are aggregated over a measurement period which typically is a month. More specifically, let the 20 years mean wind speed on day s be denoted by $w_{20}(s)$. The Nordix Index over a measurement period from τ_1 to τ_2 is then defined as

$$N(\tau_1, \tau_2) = 100 + \sum_{s=\tau_1}^{\tau_2} W(s) - w_{20}(s). \quad (4.1)$$

A day s in the measurement period with more than average wind contributes to an increased value of the index, while a lower wind speed reduces it. An index value higher than 100 means that the wind speed has over all been higher than the average over the measurement period. This means that a wind power producer had better conditions than expected. If the Nordix Index ends up below 100 over the measurement period a producer had less than average conditions for power production. A thorough description of the index can be found on the web page of the US Futures Exchange.³

The wind futures contracts at the US Futures Exchange are settled against the Nordix Index $N(\tau_1, \tau_2)$ for a specified monthly measurement period. The contract pays the money equivalent to the index value at the end of the measurement period τ_2 . We denote the futures price at time $t \leq \tau_1$ by $F(t, \tau_1, \tau_2)$. Assuming financial settlement at time τ_2 , we define the futures price using the usual definition for contracts on non-traded commodities (see e.g. Benth et al., 2008)⁴

$$F(t, \tau_1, \tau_2) \triangleq \mathbb{E}_Q[N(\tau_1, \tau_2) | \mathcal{F}_t]. \quad (4.2)$$

Here, Q is the risk-neutral probability. Hence, the futures price is the predicted risk-neutral Nordix Index value.

Let us explain the motivation behind the definition in Eq. (4.2). A probability Q is risk-neutral if it is equivalent to P and all tradeable assets in the market are martingales after discounting (see e.g. Bjørk, 1998). In a market where the underlying asset is tradeable, we can derive the futures price on that asset by the so-called buy-and-hold strategy. This strategy consists of buying the asset, and financing the purchase by a bank loan. It is well-known that then the price of the futures contract can be represented as

$$f(t, T) = S(t)e^{r(T-t)},$$

where maturity is T and the underlying asset price is $S(t)$ at time $0 \leq t \leq T$. The risk-free interest rate is $r > 0$. Under appropriate conditions on the dynamics of stochastic process $S(t)$,⁵ one can show that there exists a probability Q being equivalent to P such that the discounted asset dynamics $\exp(-rt)S(t)$ is a martingale with respect to Q , and as an implication, the futures price becomes

$$f(t, T) = \mathbb{E}_Q[S(T) | \mathcal{F}_t].$$

Observe that the futures price itself becomes a martingale with respect to Q by definition. Obviously, the wind speed is not a tradeable asset, and hence not the Nordix Index. This has the implication that we cannot use the argument of the buy-and-hold strategy to pin down the wind futures price. However, in order to ensure an arbitrage-free price dynamics, we define the futures price as in Eq. (4.2). In this way the futures price is a martingale by definition for all probabilities Q being equivalent to P . Furthermore, Q trivially becomes risk-neutral and, in particular, $Q=P$ is a valid choice of a risk-neutral probability.

A probability Q being different than P has the financial interpretation as the inclusion of a risk premium in the futures price. The risk premium is the difference between the futures price and the expected Nordix Index, and typically in commodity markets the premium is negative due to a hedging pressure among producers (prices are said to be backwardating since the producers accept a lower futures price than expected spot price at delivery, see Hull, 2000). We model the risk premium by a parametric class of probabilities Q^θ , where θ is a square integrable function on $[0, T]$, that is,

$$\int_0^T \theta^2(t) dt < \infty. \quad (4.3)$$

In the literature, this function is commonly known as the *market price of risk* and is directly related to the risk premium (see Benth et al., 2008). To introduce Q^θ , define the process B^θ as

$$B^\theta(t) = B(t) - \int_0^t \theta(s) ds. \quad (4.4)$$

By the square-integrability of θ , the Novikov condition holds, thus by Girsanov's Theorem (see e.g. Karatzas and Shreve, 1994) there exists a probability Q^θ equivalent to P such that B^θ is a Q^θ -Brownian motion on $[0, T]$.

The dynamics of $\mathbf{X}(t)$ under Q^θ becomes

$$d\mathbf{X}(t) = (A\mathbf{X}(t) + e_p \sigma(t) \theta(t)) dt + e_p \sigma(t) dB^\theta(t). \quad (4.5)$$

Further, by applying the multidimensional Itô Formula (see e.g. Karatzas and Shreve, 1994), the stochastic process $\mathbf{X}(t)$ has the explicit dynamics under Q^θ defined as

$$X(s) = \exp(A(s-t))x + \int_t^s \exp(A(s-u)) e_p \sigma(u) \theta(u) du + \int_t^s \exp(A(s-u)) e_p \sigma(u) dB^\theta(u), \quad (4.6)$$

⁵ $S(t)$ must be a semimartingale, see Benth et al. (2008).

³ www.usfe.com/products-re-wind.html.

⁴ In practice, the contract is settled financially based on a monetary value of the index. This value is usually the index times a constant. We choose this constant as one here for simplicity.

Table 4

The estimated autoregression parameters for the CAR(4) model

Parameter	α_1	α_2	α_3	α_4
Estimate	3.645	5.039	3.133	0.712

for $0 \leq t \leq s \leq T$. In particular, when $s \geq t$, $X_1(s)$ given $\mathbf{X}(t) = \mathbf{x}$ is normally distributed with mean

$$\mu_\theta(t, s, \mathbf{x}) \stackrel{\Delta}{=} e^{\int_t^s \exp(A(s-u)) e_p \sigma(u) \theta(u) du}, \quad (4.7)$$

and variance

$$\Sigma^2(t, s) \stackrel{\Delta}{=} \int_t^s \sigma^2(u) \{e^{\int_t^s \exp(A(s-u)) e_p}\}^2 du. \quad (4.8)$$

Note that under P , we find the same variance Σ^2 , but the mean will be $\mu_0(t, s, \mathbf{x})$.

We now derive an explicit relation for the wind futures price $F(t, \tau_1, \tau_2)$ based on the risk-neutral probabilities Q^θ . After slightly rewriting the futures price expression in Eq. (4.2) using the definition of the Nordix Index, we have

$$F(t, \tau_1, \tau_2) = \bar{w}_{20}(\tau_1, \tau_2) + \sum_{s=\tau_1}^{\tau_2} \mathbb{E}_\theta[W(s)|\mathcal{F}_t], \quad (4.9)$$

where, $\bar{w}_{20}(\tau_1, \tau_2)$ is 100 subtracted the aggregation of $w_{20}(s)$ over the measurement period from τ_1 to τ_2 ,

$$\bar{w}_{20}(\tau_1, \tau_2) = 100 - \sum_{s=\tau_1}^{\tau_2} w_{20}(s). \quad (4.10)$$

We use the notation $E_\theta[\cdot]$ for the expectation operator with respect to Q^θ .

To derive the futures price, we must calculate the conditional expectation of $W(s)$ given $\mathcal{F}_t$, for $s \geq t$. This is done in the following Lemma:

Lemma 4.1. Let $0 \leq t \leq s \leq T$, then for $\lambda \in (0, 1]$ it holds that

$$\mathbb{E}_\theta[W(s)|\mathcal{F}_t] = M_{\frac{1}{\lambda}} \left(1 + \lambda \left(\Lambda(s) + \mu_\theta(t, s, X(t)), \lambda^2 \Sigma^2(t, s) \right) \right),$$

where $M_k(a, b^2)$ is the k 'th moment of a normal random variable with mean a and variance b^2 . When $\lambda = 0$ we have

$$\mathbb{E}_\theta[W(s)|\mathcal{F}_t] = \exp \left(\Lambda(s) + \mu_\theta(t, s, X(t)) + \frac{1}{2} \Sigma^2(t, s) \right).$$

The functions $\mu_\theta(t, s, \mathbf{x})$ and $\Sigma^2(t, s)$ are defined in Eqs. (4.7) and (4.8), respectively,

Proof. From Eqs. (3.3) and (4.6), we have that the wind speed at time s given $\mathcal{F}_t$ can be represented (for $\lambda \neq 0$) as

$$W(s) = \left(\lambda(\Lambda(s) + \mu_\theta(t, s, X(t)) + \Sigma(t, s)Z) + 1 \right)^{1/\lambda},$$

where Z is a standard normally distributed random variable independent of $\mathcal{F}_t$. Further, $\mu_\theta(t, s, X(t))$ is $\mathcal{F}_t$ -measurable. Hence, the result follows from a direct calculation. The lognormal case $\lambda = 0$ follows similarly. $\square$

Inserting the expression obtained in Lemma 4.1 into Eq. (4.9), we get the futures price explicitly given as

$$F(t, \tau_1, \tau_2) = \bar{w}_{20}(\tau_1, \tau_2) + \sum_{s=\tau_1}^{\tau_2} M_{\frac{1}{\lambda}} \left(1 + \lambda(\Lambda(s) + \mu_\theta(t, s, X(t))), \lambda^2 \Sigma^2(t, s) \right), \quad (4.11)$$

whenever $\lambda \in (0, 1]$. For the case $\lambda = 0$, when the wind speed is the exponential of the CAR(p)-process, we have

$$F(t, \tau_1, \tau_2) = \bar{w}_{20}(\tau_1, \tau_2) + \sum_{s=\tau_1}^{\tau_2} \exp \left(\Lambda(s) + \mu_\theta(t, s, X(t)) + \frac{1}{2} \Sigma^2(t, s) \right). \quad (4.12)$$

The futures prices are explicitly depending on $\mathbf{X}(t)$, that is, the wind speeds of today and the previous $p-1$ days if considered on a discrete time scale. Thus, the price will not be Markovian. However, it is Markovian considered as a p -dimensional process in $\mathbf{X}(t)$. Recall also the assumption that $1/\lambda$ is a natural number whenever $\lambda > 0$, ruling out the possibility of, for instance, taking a root of a negative number in the calculation of $M_{1/\lambda}$.

We continue with a discussion on the volatility term structure associated to the wind futures. For this purpose, define for $t \leq s$

$$f(t, s) \stackrel{\Delta}{=} \mathbb{E}_\theta[W(s)|\mathcal{F}_t]. \quad (4.13)$$

We note that $f(t, s)$ can be considered as the price of a futures contract written on the wind speed $W(s)$ at time s . Hence, the wind futures on the Nordix Index can be looked upon as the aggregation of futures contracts on the wind speed at each day in the measurement period, and the price can be represented as

$$F(t, \tau_1, \tau_2) = \bar{w}_{20}(\tau_1, \tau_2) + \sum_{s=\tau_1}^{\tau_2} f(t, s). \quad (4.14)$$

We analyse the time dynamics, and more specifically, the volatility of the simple futures prices $f(t, s)$. The risk-neutral dynamics is given in the next Proposition.

Proposition 4.2. For $0 \leq t \leq s$ it holds

$$\frac{df(t, s)}{f(t, s)} = g_\lambda(t, s, X(t)) \{e^{\int_t^s \exp(A(s-u)) e_p \sigma(u)} dB^\theta(u)\},$$

where $g_0(t, s, X(t)) = 1$ and

$$g_\lambda(t, s, \mathbf{x}) = \frac{M_{\frac{1}{\lambda}-1} \left(1 + \lambda(\Lambda(s) + \mu_\theta(t, s, \mathbf{x})), \lambda^2 \Sigma^2(t, s) \right)}{M_{\frac{1}{\lambda}} \left(1 + \lambda(\Lambda(s) + \mu_\theta(t, s, \mathbf{x})), \lambda^2 \Sigma^2(t, s) \right)},$$

when $\lambda \in (0, 1]$.

Proof. This follows from a direct application of Itô's Formula in multi-dimensions along with the fact that

$$\frac{\partial}{\partial \mathbf{x}} M_k(\mathbf{x}, b^2) = k M_{k-1}(\mathbf{x}, b^2)$$

for $k > 1$. The latter follows from a direct calculation, and is used in the derivation with Itô's Formula. $\square$

We note that there is a significant difference between the lognormal case with $\lambda = 0$ and the other, since in the former case the volatility is deterministic. When $\lambda \neq 0$, the futures prices will have a stochastic volatility, depending explicitly on the state of $\mathbf{X}(t)$.

The term $e^{\int_t^s \exp(A(s-u)) e_p}$ is the autoregressive damping factor. Restricting our attention for a moment to the case $p = 1$, that is, the case of a CAR(1)-model (Ornstein-Uhlenbeck process), this damping factor reduces to $\exp(-\alpha_1(s-t))$. Here, α_1 is the speed of mean reversion in the Ornstein-Uhlenbeck process, and we observe an exponentially increasing factor towards one with time to maturity $s-t$ going to zero. Hence, the volatility of $f(t, s)$ is converging at an exponential rate to the wind speed volatility $\sigma(t)$ when time to maturity $s-t$ goes to zero. This feature of $f(t, s)$ is called the Samuelson effect.

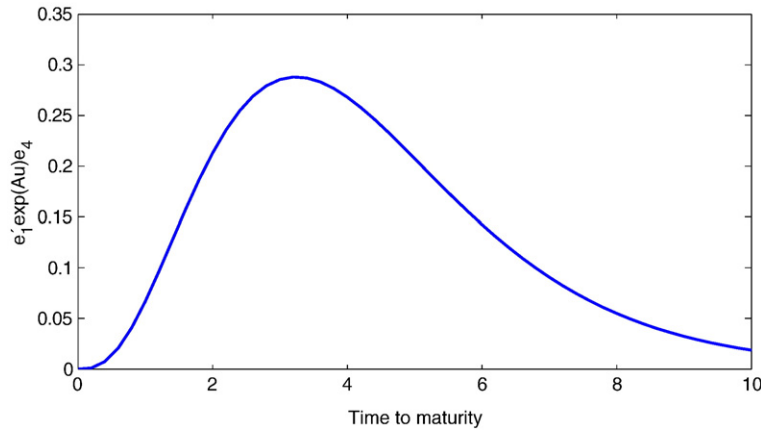


Fig. 5. $e_1' \exp(Au) e_4$ as a function of time to maturity u measured in days. The matrix A is the one estimated on data from the New York state wind farm region I.

The case of a higher-order autoregressive model ($p > 1$) is fundamentally different. Note that for $p > 1$,

$$\lim_{t \uparrow s} e_1' \exp(A(s-t)) e_p = e_1' I e_p = 0,$$

where I is the $p \times p$ identity matrix. Hence, it holds that the volatility of $f(t, s)$ tends to zero when time to maturity of the contract goes to zero. This can be explained by the fact that we have a memory in the underlying wind speed dynamics due to the higher-order autoregression, and therefore we can predict the wind speed in the near future with a high degree of certainty compared to the “memoryless” case $p = 1$. This will have the effect that there is only negligible uncertainty in the forward contract price close to maturity because we almost know the wind speed at maturity. Note that we still have

$$\lim_{t \uparrow s} f(t, s) = f(s, s) = W(s).$$

In conclusion, when the underlying process has a memory like a higher-order autoregression dynamics, the Samuelson effect is violated close to maturity since the volatility of the futures contract will decay to zero. In Fig. 5 we plot $e_1' \exp(Au) e_4$ as a function of time to maturity u , with A being the matrix estimated from the New York state data discussed above. We see that reasonably far from maturity we have values close to zero, while the function increases up to maturity, except in the last period, where it rapidly decays to zero again. Scaling this with $\sigma(t)$ will only affect the level of the function, and we therefore conclude the violation of the Samuelson effect for this model.

It is worthwhile noticing that the volatility level is seasonal, in the sense that the volatility of a specific contract depends on the time of the year it matures. A summer contract will naturally have a lower volatility than contracts in the spring and autumn, since the value of $\sigma(t)$ is bigger in these periods than in summer (see Fig. 4). This will clearly influence the value of derivatives on these futures, which also will become seasonally varying.

We next discuss the case when $\lambda \in (0, 1]$, where we have an additional scaling factor g_λ as defined in the Prop. 4.2. We observe that since $\mathbf{X}(t)$ is a CAR(p)-process driven by a Brownian motion, it may attain values on the whole real line, including negative values. This means that the function g_λ may become negative. In fact, we may have situations where it changes sign over time, and therefore explodes. This becomes clear when we look at the limit with time to maturity approaching zero, where it holds

$$\lim_{t \uparrow s} g_\lambda(t, s, x) = \frac{1}{1 + \lambda(A(s) + X_1(s))}.$$

Note that this is a problem only as long as we insist on defining the volatility for this class of processes. The time-dynamics of the forward itself is well-behaved, since it is given by

$$df(t, s) = M_{\lambda-1} \left(1 + \lambda(A(s) + \mu_\theta(t, s, X(t))) \right) \lambda^2 \Sigma^2(t, s) \sigma(t) e_1' \exp(A(s-t)) e_p dB^\theta(t).$$

Since we assumed that $1/\lambda$ is a natural number and $\lambda \leq 1$, $M_{1/\lambda-1}$, the $1/\lambda-1$ moment of a normal random variable is well-defined. Note however, that it may be negative, leading to a negative forward price. As already remarked earlier, this is not a big problem from a practical point of view since the probability for this to happen will be very low (at least for the fitted model for New York).

Finally, we make some notes on the Nordix Index futures. There is no simple volatility expression for these contracts, as can be seen from the dynamics of f . However, since the futures price $F(t, \tau_1, \tau_2)$ can be viewed as an aggregation of the prices $f(t, s)$ for s in the measurement period, one may think of the volatility of F as an aggregation of the volatilities of f . This may be highly speculative, but in electricity markets similar approximations have been used with success (see e.g. Bjerkund et al., 2000; Benth et al., 2008). This means that also the volatility of Nordix Index futures will violate the Samuelson effect. Due to the averaging over the measurement period, we will not see a volatility that collapses to zero at maturity. However, the volatility will increase when time approaches τ_1 , except in last few days where the volatility will flatten and even drop down. Especially for futures with short measurement period the volatility will drop down in the last trading days before the measurement period starts, whereas contracts with longer measurement periods will have a flattening volatility. These considerations are adopted from temperature futures in Benth et al. (2007).

4.1. Pricing of options on wind futures

We end this section with a discussion on the pricing of options written on the futures discussed above.

Consider a European call option written on a wind futures with measurement period $[\tau_1, \tau_2]$. We suppose that the exercise time of the option is $\tau < \tau_1$, and strike price is set to $K > 0$. From the theory of option pricing (see e.g. Bjørk, 1998), the arbitrage-free price of this option is

$$C(t, \tau, K, \tau_1, \tau_2) = e^{-r(\tau-t)} \mathbb{E}_\theta[\max(F(\tau, \tau_1, \tau_2) - K, 0) | \mathcal{F}_t]. \quad (4.15)$$

Here, $t \leq \tau$ and r is the risk-free interest rate.

From the explicit formula of the wind futures price in Eq. (4.11), we have

$$F(\tau, \tau_1, \tau_2) = \bar{w}_{20}(\tau_1, \tau_2) + \sum_{s=\tau_1}^{\tau_2} M_{\lambda} \left(\left(1 + \lambda(\Lambda(s) + \mu_{\theta}(t, s, X(\tau))) \right), \lambda^2 \Sigma^2(t, s) \right),$$

for $\lambda \in (0, 1]$, and from Eq. (4.12)

$$F(\tau, \tau_1, \tau_2) = \bar{w}_{20}(\tau_1, \tau_2) + \sum_{s=\tau_1}^{\tau_2} \exp \left(\Lambda(s) + \mu_{\theta}(\tau, s, X(\tau)) + \frac{1}{2} \Sigma^2(\tau, s) \right),$$

for $\lambda = 0$. Recall the explicit risk-neutral dynamics of $\mathbf{X}(s)$ in (4.6), and insert this into $\mu_{\theta}(\tau, s, \mathbf{X}(\tau))$. Conditioning on $\mathbf{X}(t)$ leads to

$$\begin{aligned} \mu_{\theta}(\tau, s, X(\tau)) &= e_1' \exp(A(s-\tau)) X(\tau) + \int_{\tau}^s \{e_1' \exp(A(s-u)) e_p\} \sigma(u) \theta(u) du \\ &= e_1' \exp(A(s-t)) X(t) + e_1' \exp(A(s-\tau)) \int_t^{\tau} \exp(A(\tau-u)) e_p \sigma(u) \theta(u) du \\ &\quad + \int_{\tau}^s \{e_1' \exp(A(s-u)) e_p\} \sigma(u) \theta(u) du \\ &\quad + e_1' \exp(A(s-\tau)) \int_t^{\tau} \sigma(u) \exp(A(\tau-u)) e_p dB^{\theta}(u) \\ &= \mu_{\theta}(t, s, X(t)) + \int_t^{\tau} \sigma(u) e_1' \exp(A(s-u)) e_p dB^{\theta}(u). \end{aligned}$$

Note that the Itô integral defines a centered normal distribution with variance

$$V^2(t, \tau, s) = \Delta \int_t^{\tau} \sigma^2(u) \{e_1' \exp(A(s-u)) e_p\}^2 du. \quad (4.16)$$

Hence, we have

$$\mu_{\theta}(\tau, s, X(\tau)) = \mu_{\theta}(t, s, X(t)) + V(t, \tau, s)Z, \quad (4.17)$$

where the equality is in distribution and Z is a standard normally distributed random variable.

Based on Eq. (4.17), introduce the function $h_{\lambda} : \mathbb{R}^{p+1} \mapsto \mathbb{R}_+$ defined as

$$\begin{aligned} h_{\lambda}(x, z) &= \bar{w}_{20}(\tau_1, \tau_2) \\ &\quad + \sum_{s=\tau_1}^{\tau_2} M_{\lambda} \left(1 + \lambda(\Lambda(s) + \mu_{\theta}(t, s, x) + V(t, \tau, s) \times z), \lambda^2 \Sigma^2(\tau, s) \right). \end{aligned} \quad (4.18)$$

for $\lambda \in (0, 1]$, and

$$\begin{aligned} h_0(x, z) &= \bar{w}_{20}(\tau_1, \tau_2) \\ &\quad + \sum_{s=\tau_1}^{\tau_2} \exp \left(\Lambda(s) + \mu_{\theta}(t, s, x) + V(t, \tau, s) \times z + \frac{1}{2} \Sigma^2(\tau, s) \right), \end{aligned} \quad (4.19)$$

for $\lambda = 0$. Then we can write the price of the call option as

$$C(t, \tau, K, \tau_1, \tau_2) = e^{-r(\tau-t)} \mathbb{E}_{\theta} [\max(h_{\lambda}(x, Z) - K, 0)]_{|x=X(t)}. \quad (4.20)$$

This representation is suitable for numerical integration or efficient Monte Carlo evaluation since we only have to find the expectation of a function of a standard normal random variable Z . The simulation of Z can be performed by known techniques (see e.g. Glasserman, 2000). Given an outcome z from Z , we must calculate $\max(h_{\lambda}(x, z) - K, 0)$, which means to perform a summation over moments of order $1/\lambda$ of a normal distribution. The expectation and variance of this normal distribution is given explicitly after numerical integration of the market price of risk function θ and the wind speed volatility σ . The moments are then known analytically. These operations can be performed efficiently on a computer.

We note that a call option on $F(\tau, \tau_1, \tau_2)$ can be considered as an average-type derivative, since from Eq. (4.14) we have

$$F(\tau, \tau_1, \tau_2) = \bar{w}_{20}(\tau_1, \tau_2) + \sum_{s=\tau_1}^{\tau_2} f(\tau, s).$$

Hence, a call on $F(\tau, \tau_1, \tau_2)$ with strike K is equivalent to a call on the sum of $f(\tau, s)$ over the maturities s from τ_1 to τ_2 , with strike $K - w_{20}(\tau_1, \tau_2)$. In

a sense we can consider this as a calendar option since it involves an option on a strip of forward contracts maturing at different calendar times.

For the degenerate case $\tau_1 = \tau_2$, we can indeed obtain an analytical option pricing formula for $\lambda = 0$. We consider the equivalent case of pricing a call option written on $f(t, s)$ with exercise time $\tau < s$ and strike K . Hence, we must calculate

$$C(t, \tau, s, K) = e^{-r(\tau-t)} \mathbb{E}_{\theta} [\max(f(\tau, s) - K, 0) | \mathcal{F}_t], \quad (4.21)$$

with $t \leq \tau$. The price is stated in the next Proposition.

Proposition 4.3. Assume $\lambda = 0$, that is, the wind speed has an exponential dynamics. The price of a call option on $f(t, s)$ with exercise time $\tau < s$ and strike K is given as

$$C(t, \tau, s, K) = e^{-r(\tau-t)} \{f(t, s) \Phi(d_1) - K \Phi(d_2)\}.$$

Here, Φ is the cumulative standard normal distribution function, $d_1 = d_2 + V(t, \tau, s)$ and

$$d_2 = \frac{\ln(f(t, s)/K) - \frac{1}{2} V^2(t, \tau, s)}{V(t, \tau, s)}.$$

The integrated volatility $V(t, \tau, s)$ is defined in Eq. (4.16).

Proof. Since $\lambda = 0$, we have from Lemma 4.1 the following risk-neutral dynamics

$$f(\tau, s) = \exp \left(\Lambda(s) + \mu_{\theta}(\tau, s, X(\tau)) + \frac{1}{2} \Sigma^2(\tau, s) \right).$$

From the derivations above, we find (in distribution) that

$$\mu_{\theta}(\tau, s, X(\tau)) = \mu_{\theta}(t, s, X(t)) + V(t, \tau, s)Z$$

with Z a standard normal random variable. Hence, (in distribution),

$$f(\tau, s) = f(t, s) \exp \left(V(t, \tau, s)Z + \frac{1}{2} \Sigma^2(\tau, s) - \Sigma^2(t, s) \right).$$

But,

$$\Sigma^2(\tau, s) - \Sigma^2(t, s) = -V(t, \tau, s),$$

and therefore (in distribution) it holds that

$$f(\tau, s) = f(t, s) \exp \left(V(t, \tau, s)Z - \frac{1}{2} V^2(t, \tau, s) \right).$$

Inserting this into the conditional expectation in Eq. (4.21) and calculating using standard properties of the normal distribution, we obtain the expression for $C(t, \tau, s, K)$ in the Proposition. $\square$

We recognize the expression for $C(t, \tau, s, K)$ as the famous Black-76 Formula for time-dependent volatilities. Black (1976) derived this formula for options on futures following a geometric Brownian motion. See e.g. Benth et al. (2008) for the formula with time-dependent volatilities. Finally, note that the option market is complete since the futures dynamics is based on a fixed Brownian motion B_{θ} . The hedging strategy for the call option written on $f(t, s)$ as analyzed above can be perfectly replicated by a delta hedge.

5. Concluding remarks

Wind power producers are exposed to significant risk from changing wind conditions. Too low wind speeds obviously reduces the production, but also changing directions of wind, gusts and excessive speeds can harm the production. In the summer 2007 the US Futures Exchange

launched a series of futures contracts written on wind speed indices in wind farm areas in New York and Texas states. These contracts are tailor-made for wind power producers to hedge their wind speed risk, the obvious major factor determining production. Further, there are call and put options written on the wind futures offering more flexibility in securing profits from wind power production. To understand the pricing of such futures and options is important for constructing efficient hedging strategies. The key input is a statistically sound wind speed model which is analytically tractable. This is what we provide in our study.

Based on the empirical findings in Šaltytė Benth and Benth (2008) we argue for a fourth-order autoregressive model for power transformed wind speed data. The data are daily wind speeds collected from three wind farm regions in New York state. A clear seasonality in the volatility of the residuals is accounted for. A similar seasonal volatility is observed in surface air temperature dynamics.

We apply our wind speed dynamics to price financial futures contracts on a wind speed index. The index is based on the aggregated deviation from a 20 years daily average over a given measurement period. Our model allows for explicit futures price dynamics expressed in terms of the moments of a normal distribution. Interestingly, the volatility function of the wind futures price violates the Samuelson effect even though the underlying model is mean-reverting. Due to the high-order autoregression, the wind speed will possess a memory which means that when we are close to maturity of the contract we can already predict with a high degree of certainty the wind in the near future. This lowers the uncertainty of the payoff from the futures contract, and hence the volatility. We predict a flattening, or even a drop in volatility close to maturity. This has impact on risk management using wind futures, and pricing of derivatives like options on wind futures.

An analysis of call options on wind futures is included, and it is shown how to do this efficiently by simulation methods. Furthermore,

we show the link to the Black-76 formula via options on futures which do not have a measurement period but rather “delivers” wind at a certain time.

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